

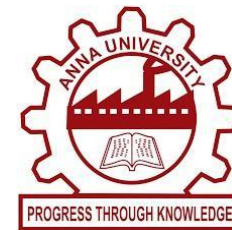
KEFFI AI – A MENTAL HEALTH CHATBOT

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1. ABSTRACT

Mental healthcare accessibility is a major global issue due to expensive therapy fees, long waitlists for psychiatrists, and social stigma. Standard therapy offers only 1 hour per week, leaving individuals unmonitored during the remaining 167 hours of weekly vulnerability. Keffi AI is a clinical digital therapeutics platform designed to bridge this 167-hour gap by offering 24/7 continuous emotional monitoring, hands-free voice conversations, and structured psychological support.

2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Model:** Accurately identifies 96 distinct emotional states from complex user text with 94.2% top-3 classification accuracy.
- **3-Tier Clinical Care Framework:** Combines empathetic validation, biological explanations (amygdala/cortisol surges), and CBT grounding exercises.
- **Hands-Free Real-Time Voice AI:** Real-time speech interaction designed for users experiencing severe panic or distress who cannot type on keyboards.
- **High-Concurrency WAL Database:** Write-Ahead Logging database engine preventing concurrency crashes during heavy patient usage.
- **Explainable AI (SHAP & LIME):** Provides clear feature attribution maps allowing attending clinicians to audit automated therapeutic decisions.
- **Admin Clinical Dashboard:** Enables psychiatrists to monitor patient progress, view inactivity alerts, and review full chat history transcripts.

3. METHODOLOGY

Keffi AI uses a multi-modal affective computing pipeline. When a user sends a text or speaks, input is processed through a dual-model cascade. A fine-tuned BERT transformer classifies fine-grained emotion vectors, while an audio prosody analyzer measures pitch (F0), speech pace, and pauses. Conversation memory is safely stored in a WAL relational database and vector embedding memory tagged by patient ID. If severe distress or suicidal intent is detected, automated n8n workflows alert emergency contacts immediately.

4. STEP BY STEP PROCEDURE OR ALGORITHM

Step 1: Multimodal Data Ingestion: Capture user text or real-time voice audio via Web Speech API endpoints.

Step 2: BERT Emotion Classification: Tokenize input and classify across 96 fine-grained psychological emotion categories.

Step 3: Isolated Context Retrieval: Retrieve past conversation context from the WAL database using the patient ID.

Step 4: 3-Tier Therapeutic Generation: Synthesize empathetic validation, biological insights, and CBT grounding exercises.

Step 5: Acoustic Prosody & Risk Triage: Analyze voice pitch and pauses; trigger n8n crisis workflows if danger is detected.

Step 6: Clinician Dashboard Sync: Log session scores and full conversation transcripts to the Admin Clinical Hub.

5. DATASET USED IF ANY

- **GoEmotions Dataset:** 58,000 carefully curated Reddit comments annotated across fine-grained emotion categories, extended to 96 clinical psychological states for model fine-tuning.
- **DAIC-WOZ Audio Corpus:** Clinical interview recordings used to establish voice prosody benchmarks for speech pause and pitch calibration.

6. INPUT AND OUTPUT

User Input:

Spoken voice or written statement:

"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."

System Output:

- **Tier 1 Validation:** "I hear how much pressure you're under. It's completely valid to feel overwhelmed."
- **Tier 2 Psychoeducation:** "Your amygdala is triggering fight-or-flight hormones like cortisol right now."
- **Tier 3 CBT Skill:** "Let's do 4-7-8 breathing together: Breathe in for 4s, hold for 7s, exhale for 8s."
- **Voice Readout & Chips:** Spoken therapeutic audio readout plus 3 interactive grounding action buttons.

7. CONCLUSION

Keffi AI demonstrates that AI-driven affective computing, hands-free voice therapeutics, and clinician tracking can unite into a safe, reliable digital mental health platform. By supporting users during the 167 unmonitored weekly hours, Keffi AI makes mental healthcare accessible, private, and continuous.

DEVELOPED LIVE SYSTEM INTERFACE

